

RELATORE

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STUDENTE

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Programmazione ad agenti in Repast e una sua applicazione al Q-Learning



UNIVERSITÀ
DELLA
CALABRIA

DIPARTIMENTO DI INGEGNERIA
INFORMATICA, MODELLISTICA,
ELETTRONICA E SISTEMISTICA

DIMES

A.A.
2023/24

OUTLINE

01

Programmazione
ad agenti

02

Repast Symphony

03

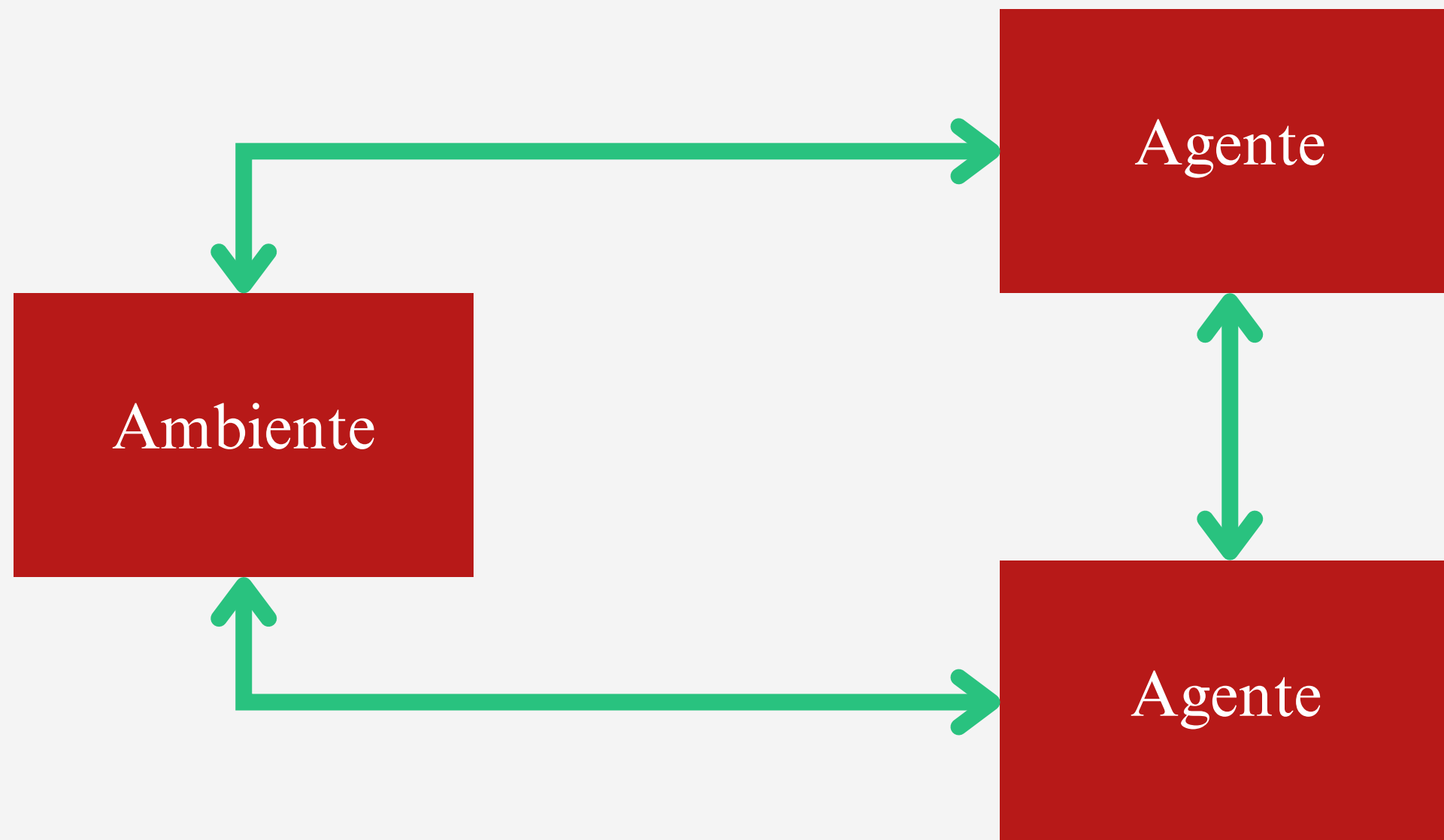
Reinforcement
Learning

04

Tic Tac Toe

PROGRAMMAZIONE AD AGENTI

01



Caratteristiche degli Agenti

Autonomia

Mobilità

Socialità

Agenti Deliberativi

Agenti Reattivi

Agenti Ibridi

Agenti del Reinforcement
Learning

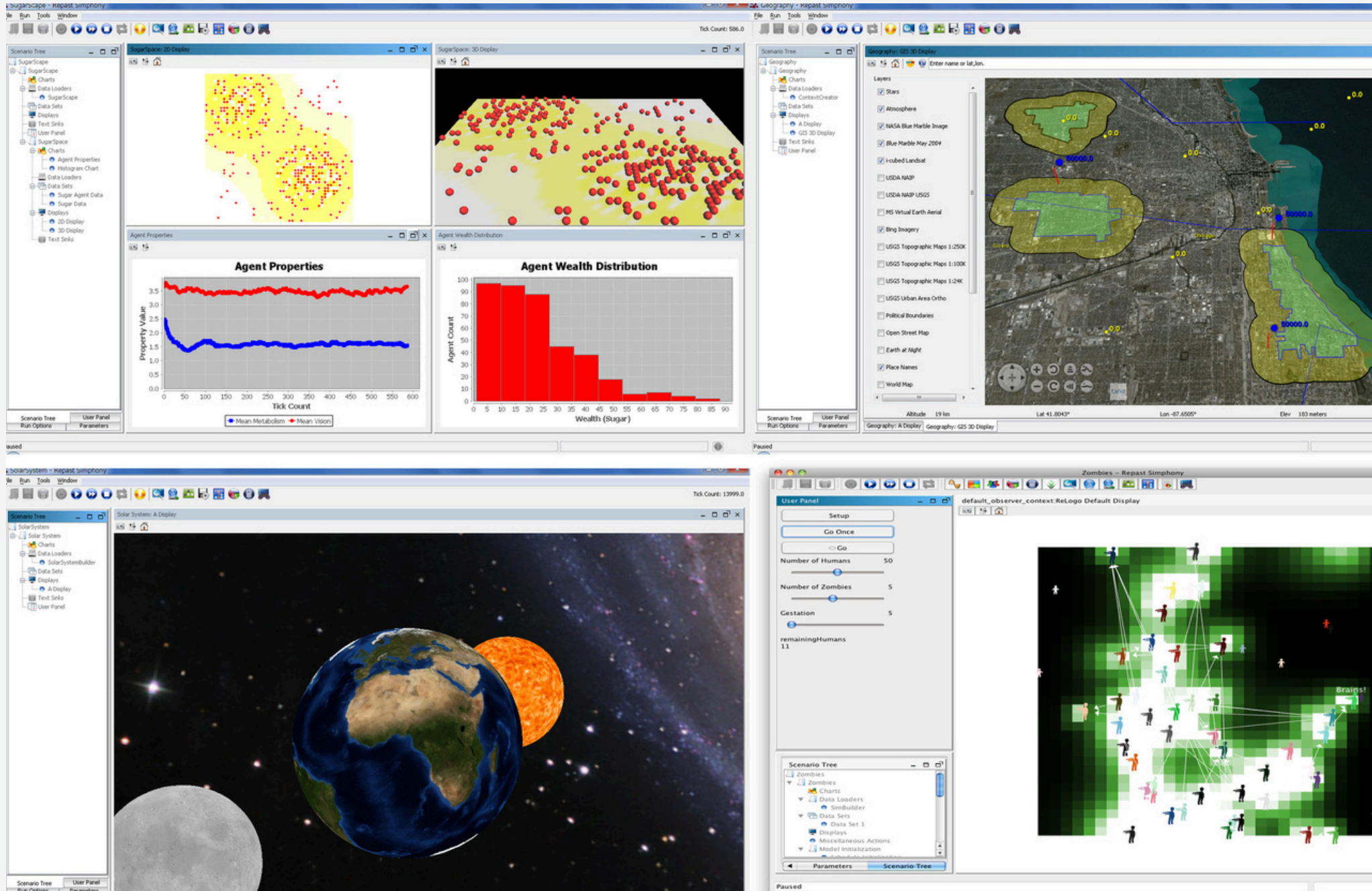
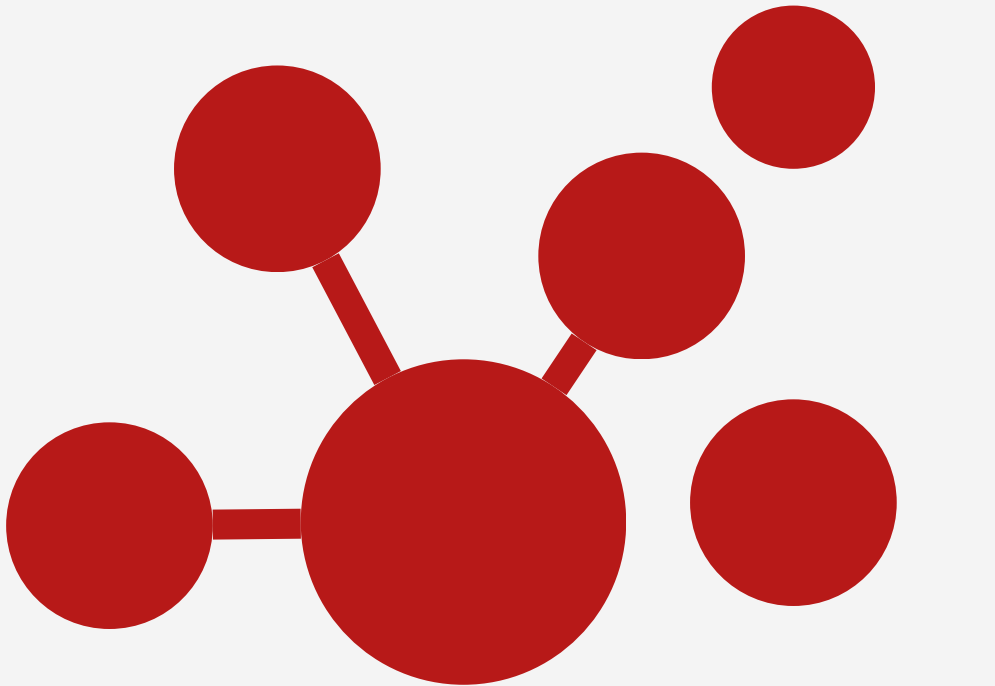
PROGRAMMAZIONE AD AGENTI

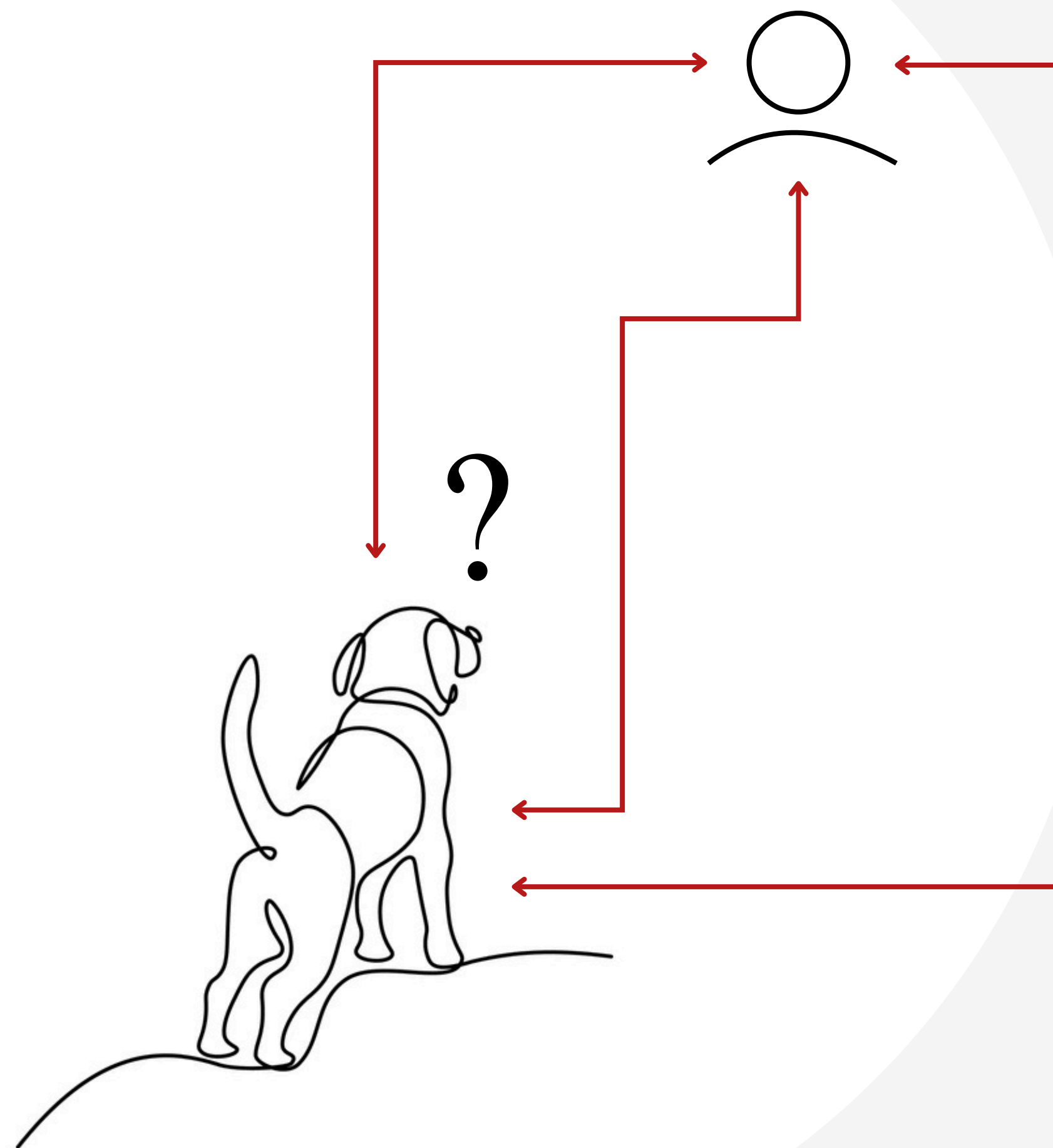
Tipologie di Agenti Intelligenti

01

REPAST SIMPHONY

Tool java in ambiente Eclipse





REINFORCEMENT LEARNING

Cos'è e come funziona?

03

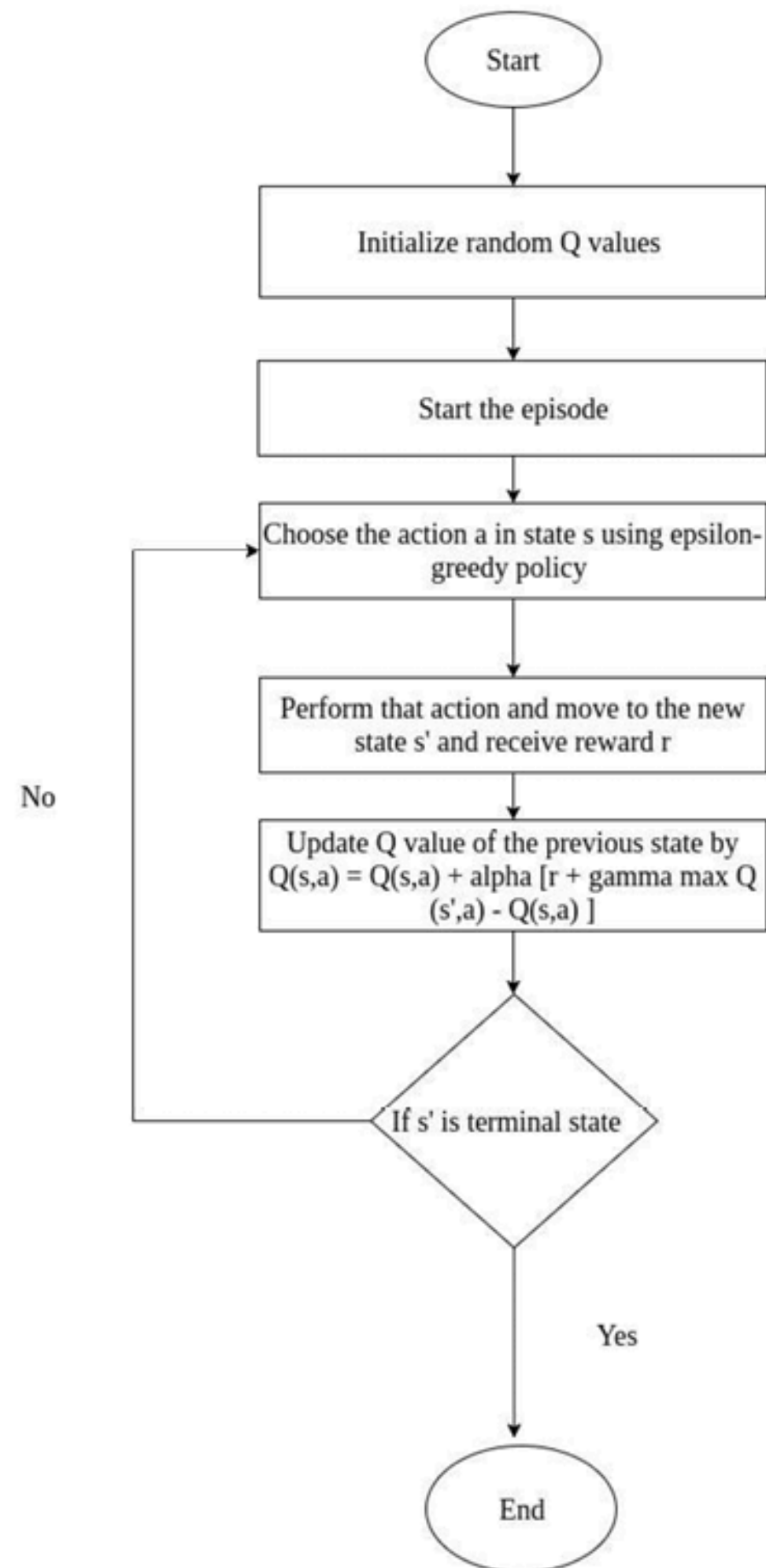


REINFORCEMENT LEARNING

L'algoritmo del Q-Learning

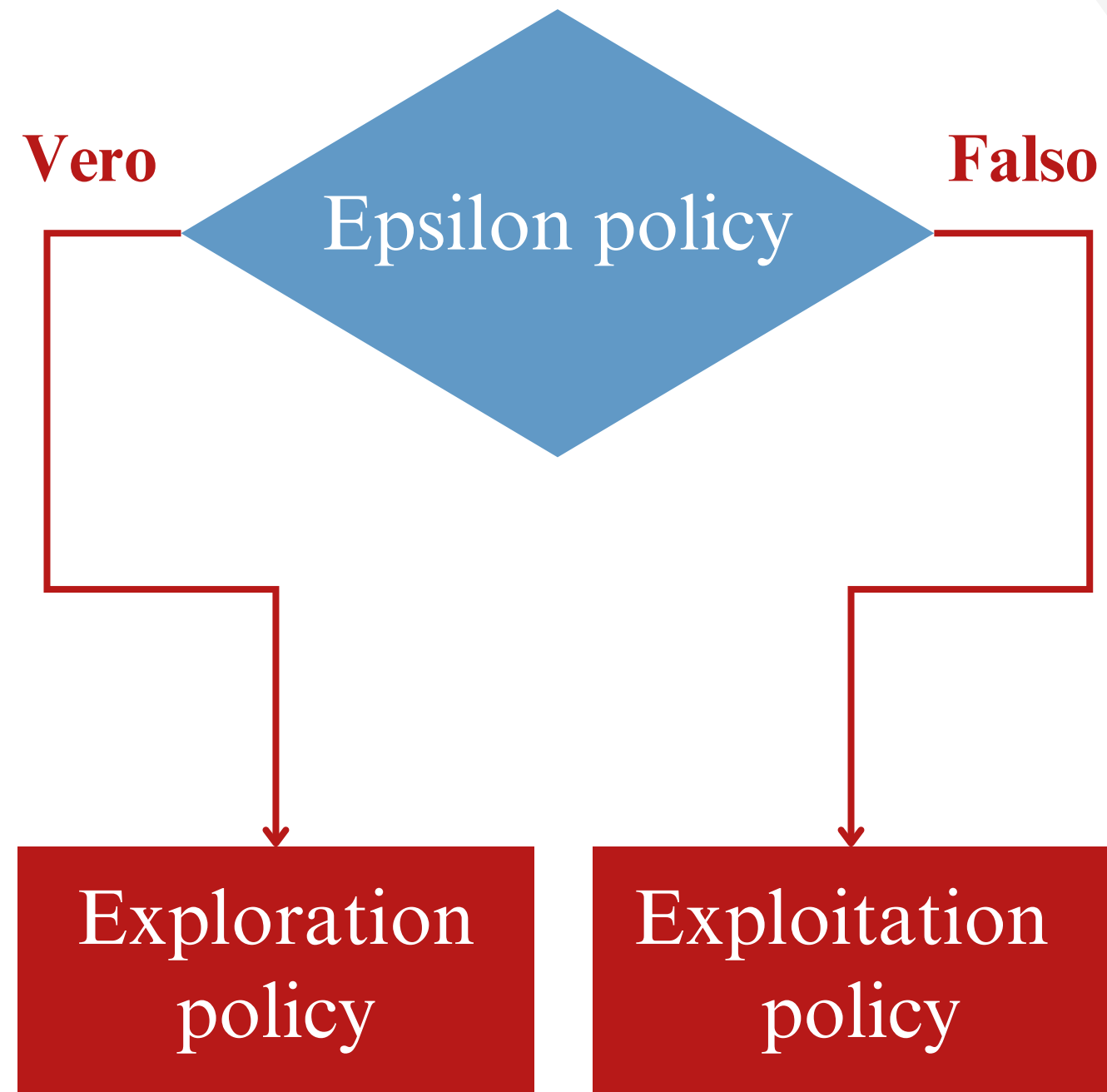
03

Q-table



Stato	Azione	Valore
Stato1	Azione1	0,88
Stato1	Azione2	0,5
Stato2	Azione3	-0,2

TUNING



REINFORCEMENT LEARNING

Epsilon - Greedy policy

03

REINFORCEMENT LEARNING

Formula di aggiornamento della Q-table

$$Q(s,a) = Q(s,a) + \alpha(r + \gamma \max_{a'} Q(s',a') - Q(s,a))$$

TUNING

s → state

a → action

r → rewards

α → learning rate

γ → discount-factor

ESEMPIO DI Q-TABLE

Stato	Azione	Valore
		0.3
		0.9
		-0.2

Griglia 3x3 con logica a passo
doppio e Epsilon fisso

Griglia 3x3 con logica a passo
singolo e Epsilon con decadimento

Griglia 3x3 con logica a passo
doppio e Epsilon con decadimento

Griglia 4x4 con 4 o 3 elementi
in fila per vincere

Interfaccia grafica Giocatore
umano Vs Agente

TIC-TAC-TOE

Applichiamo il Q-Learning con Repast

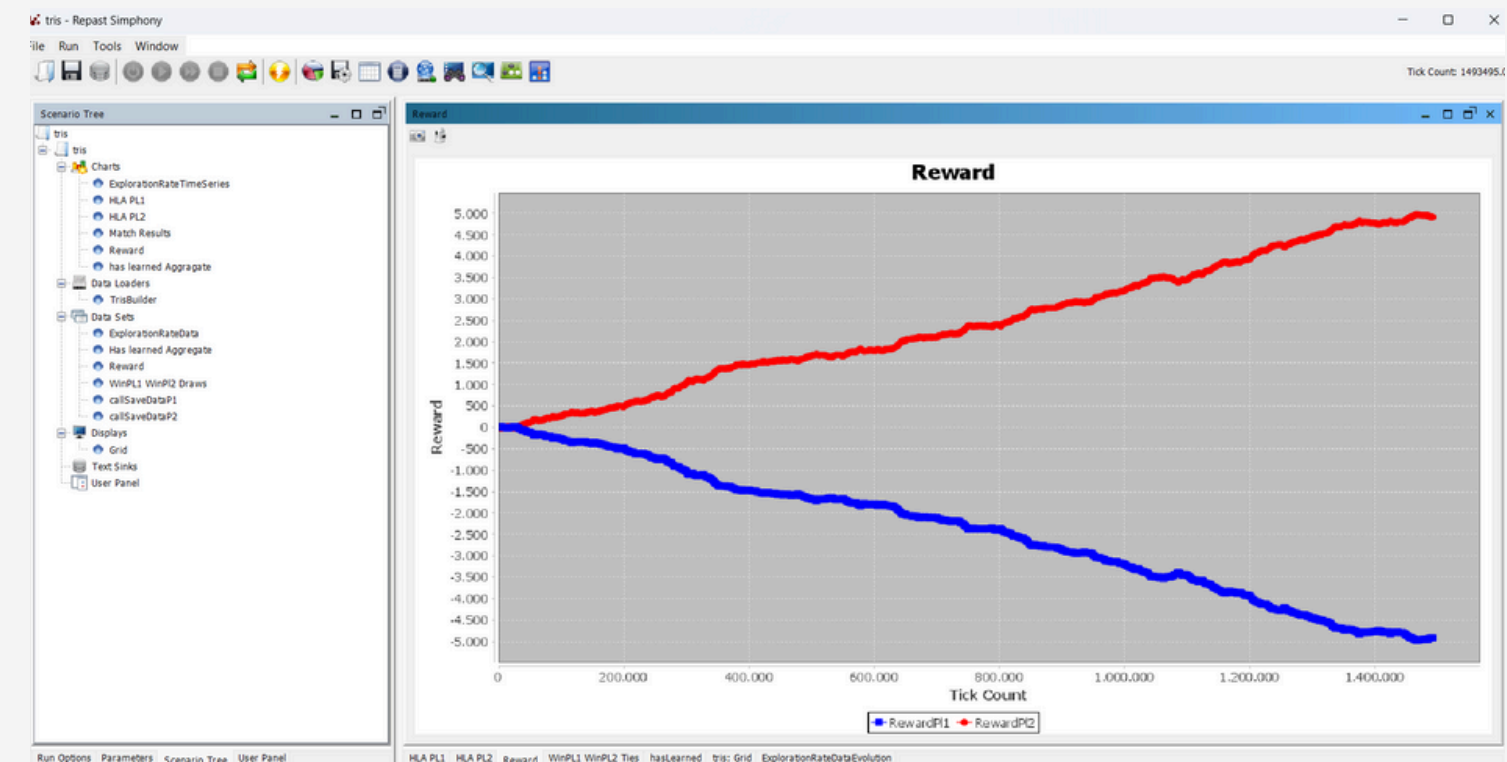
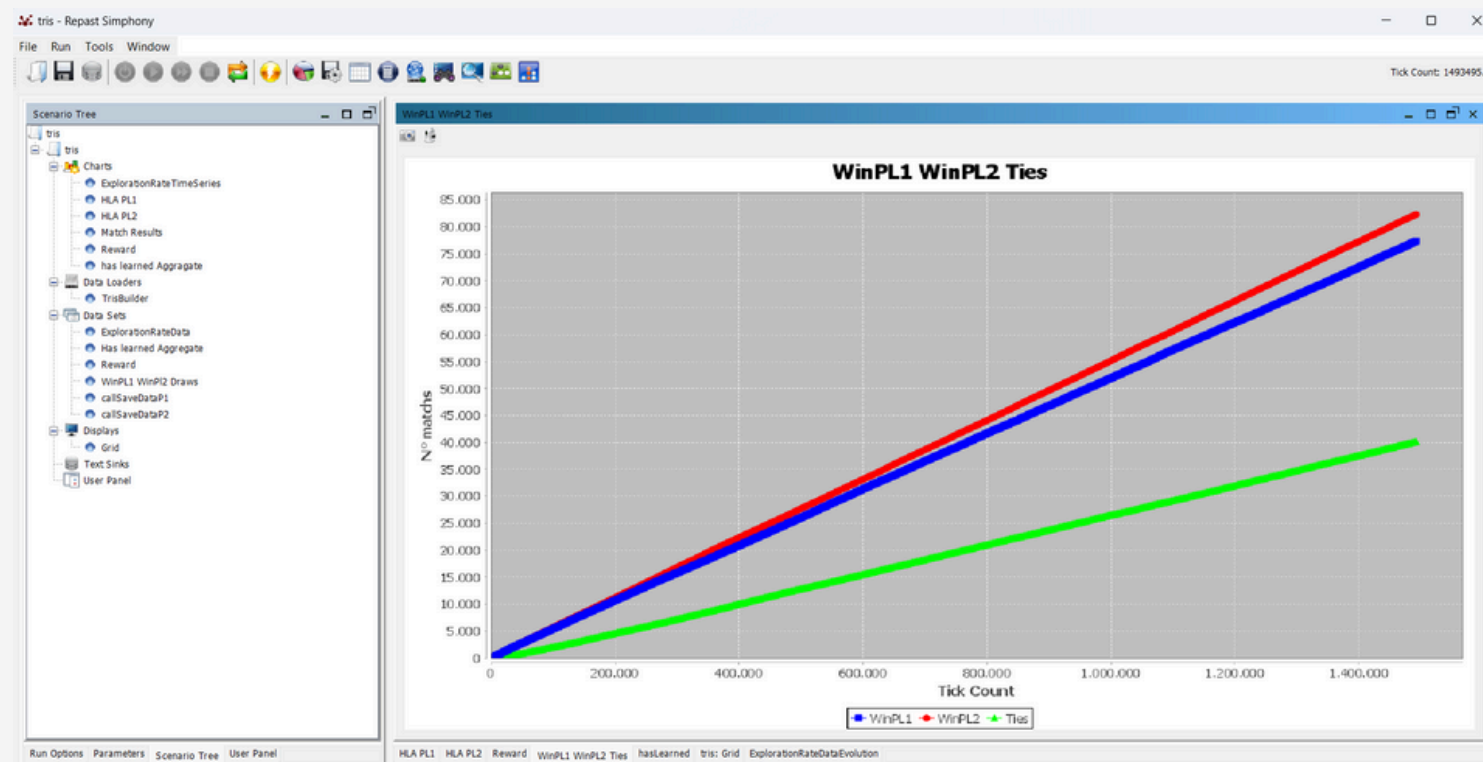
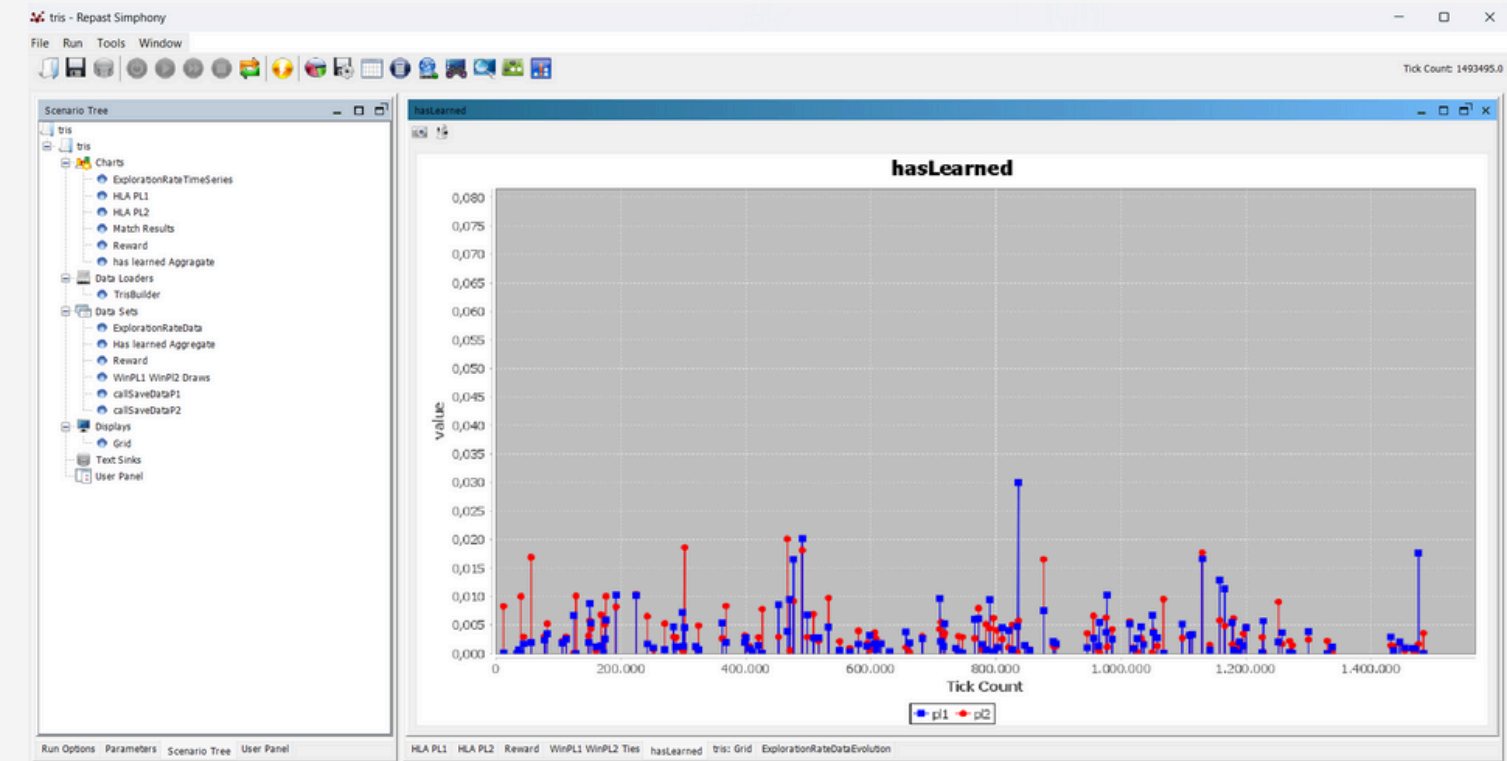
04



TIC-TAC-TOE

04

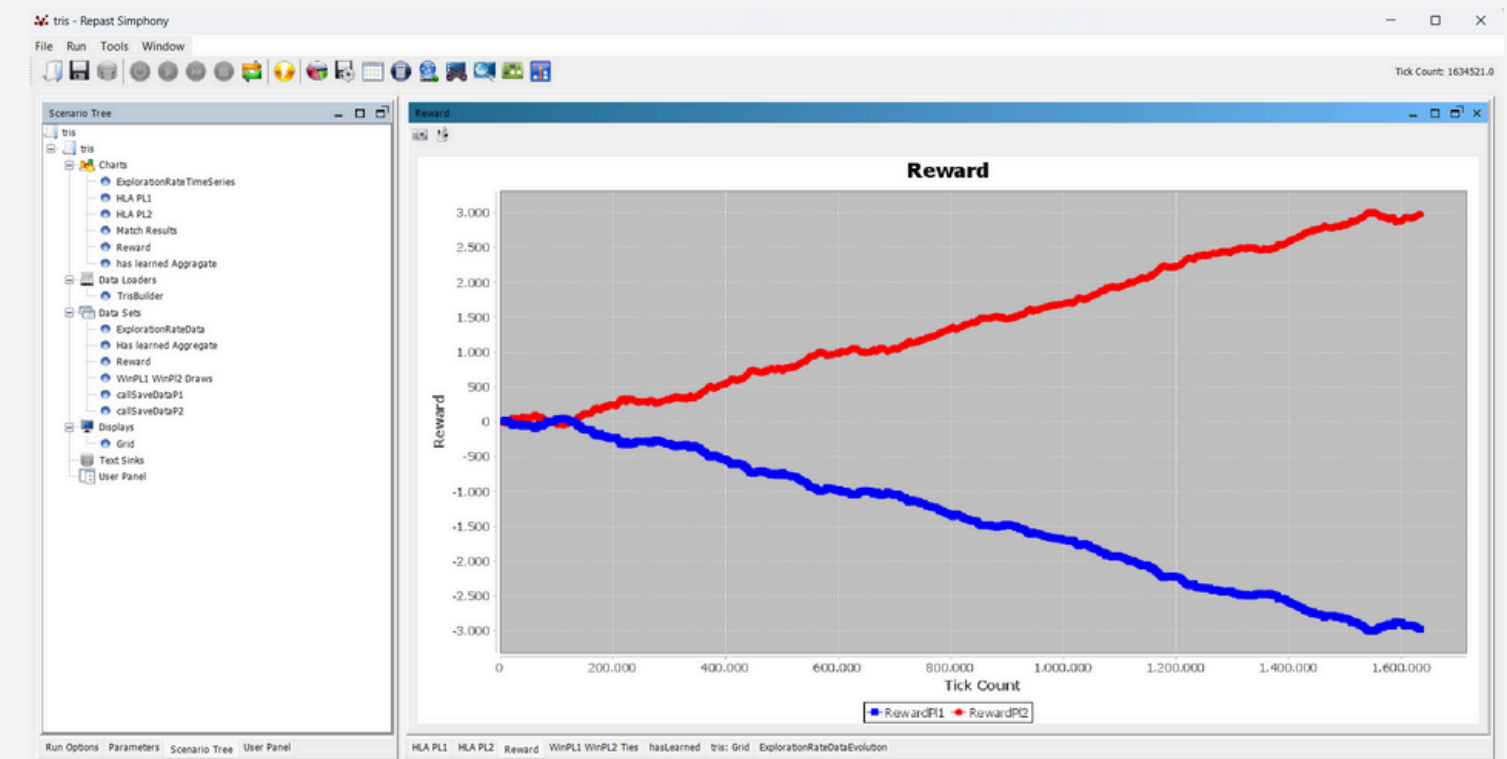
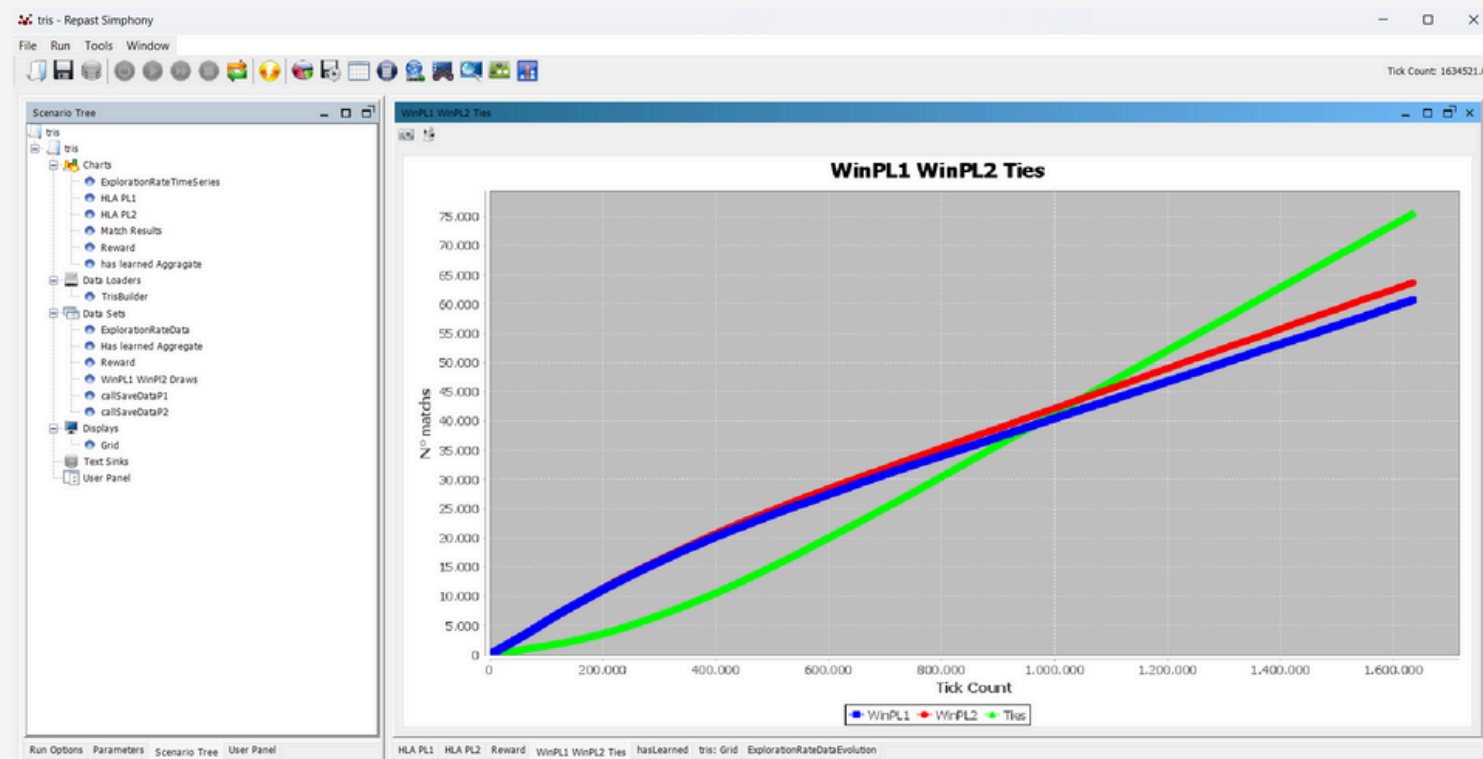
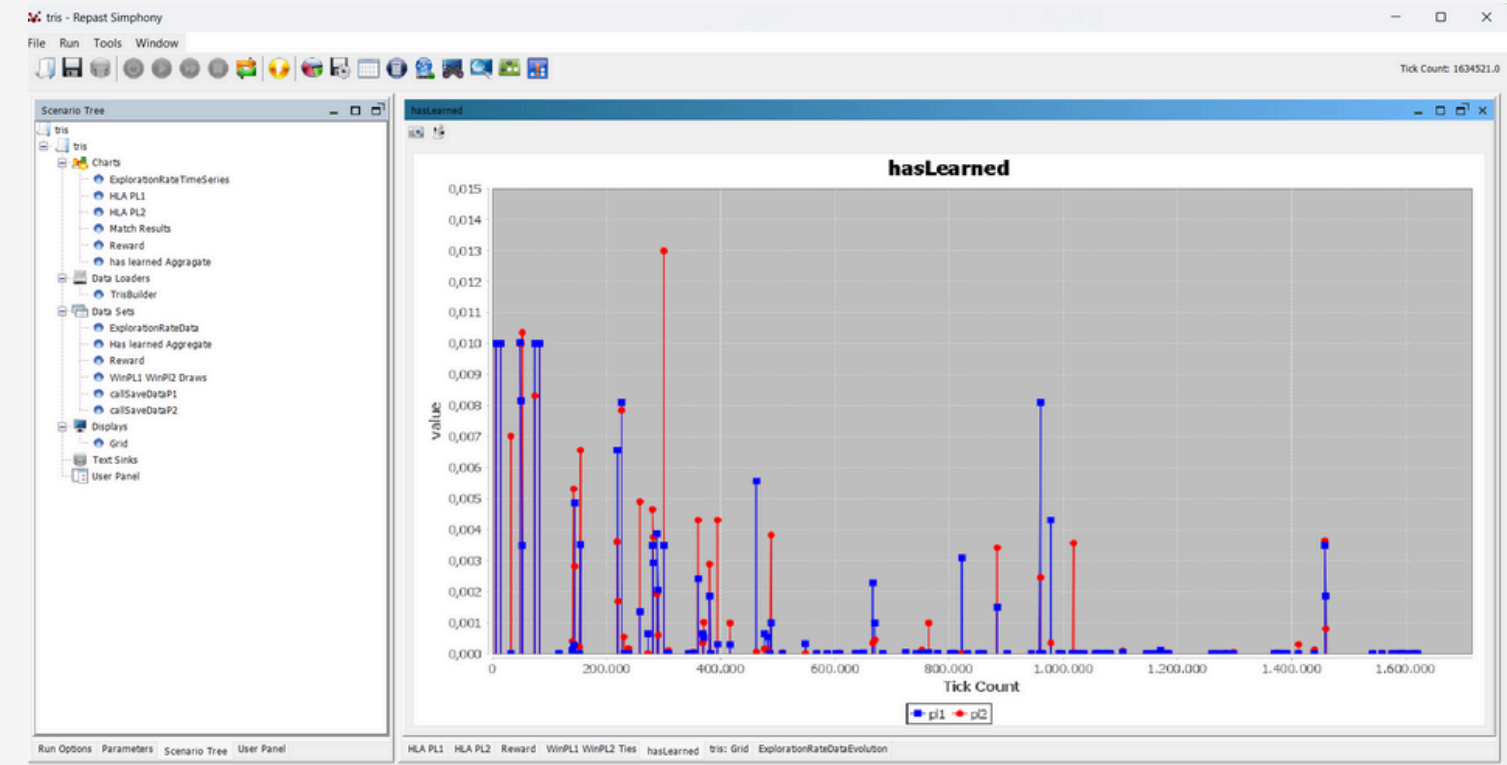
Griglia 3x3 con logica a
passo doppio e Epsilon fisso



TIC-TAC-TOE

04

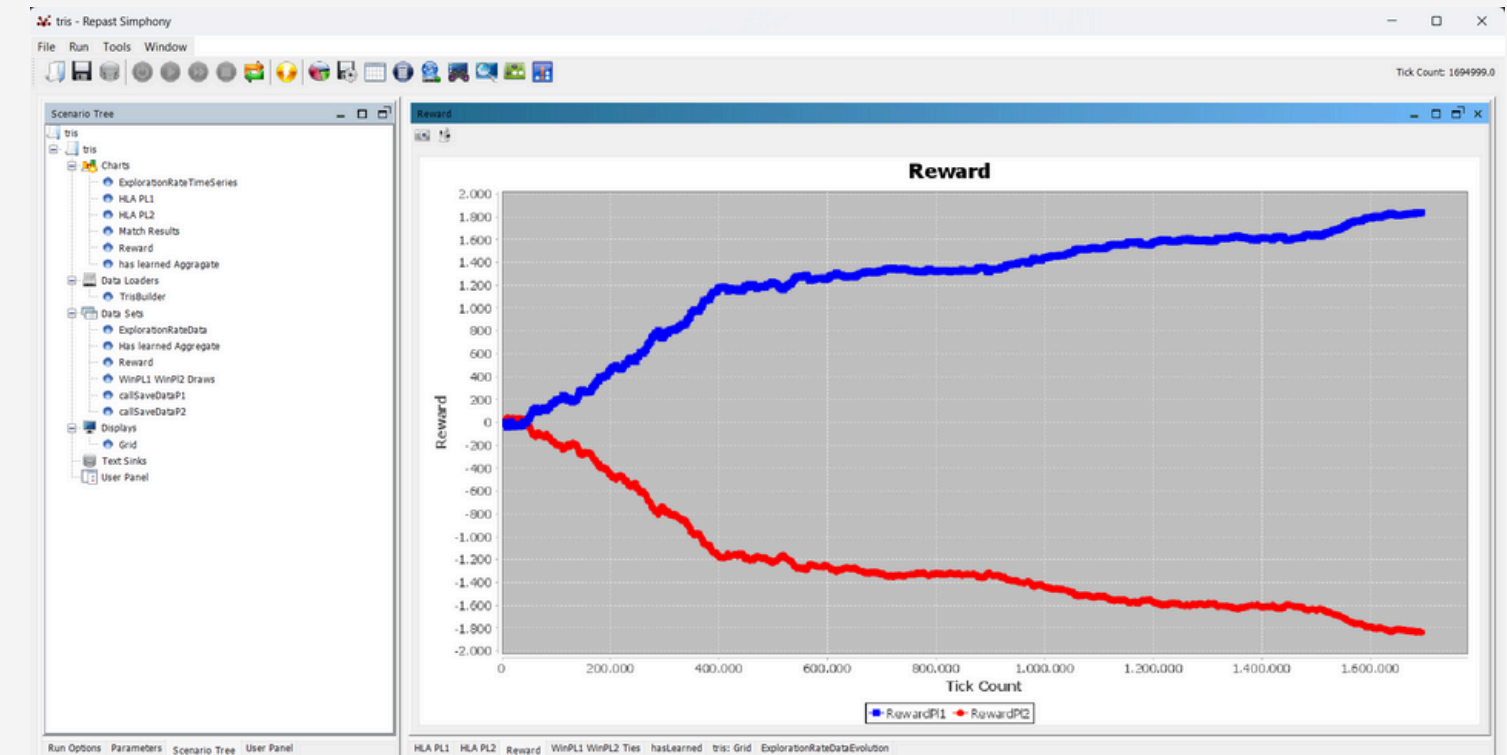
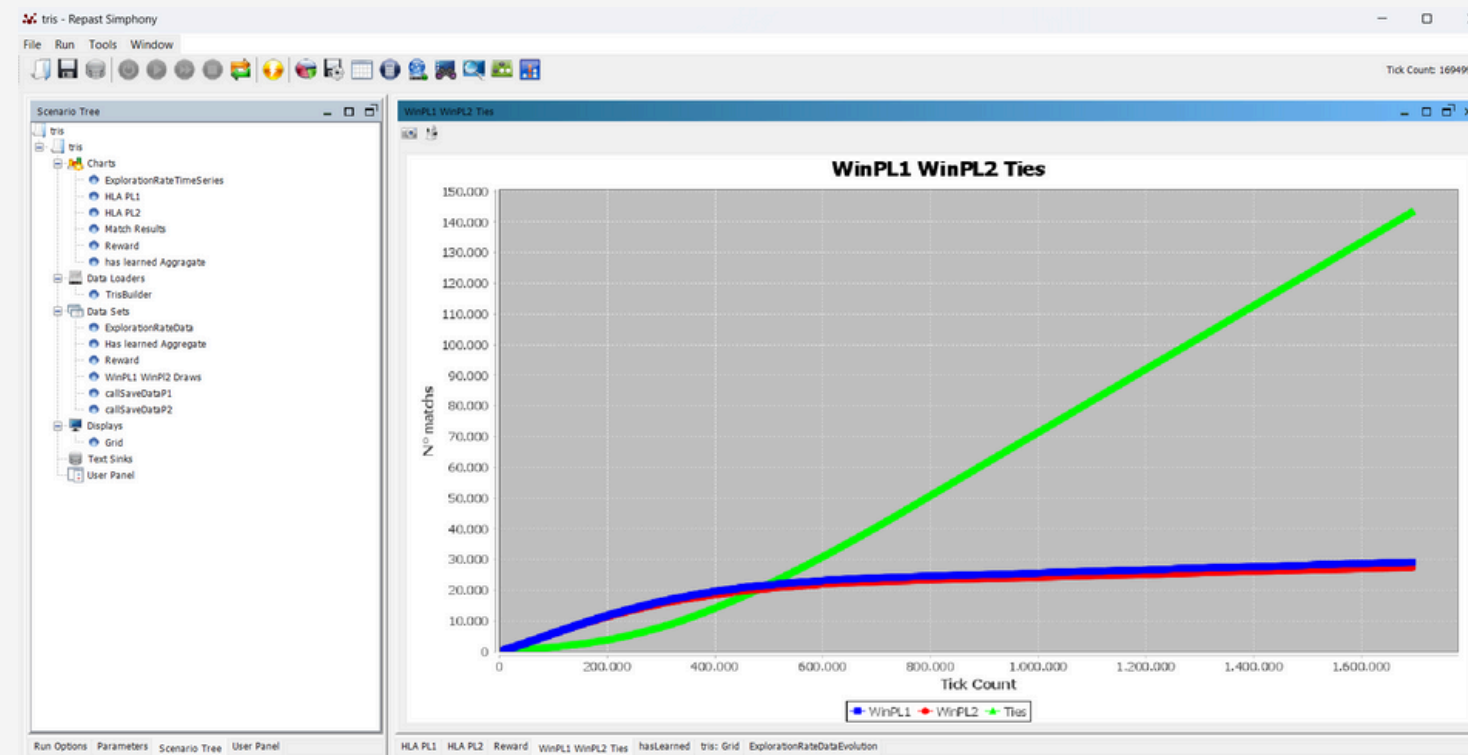
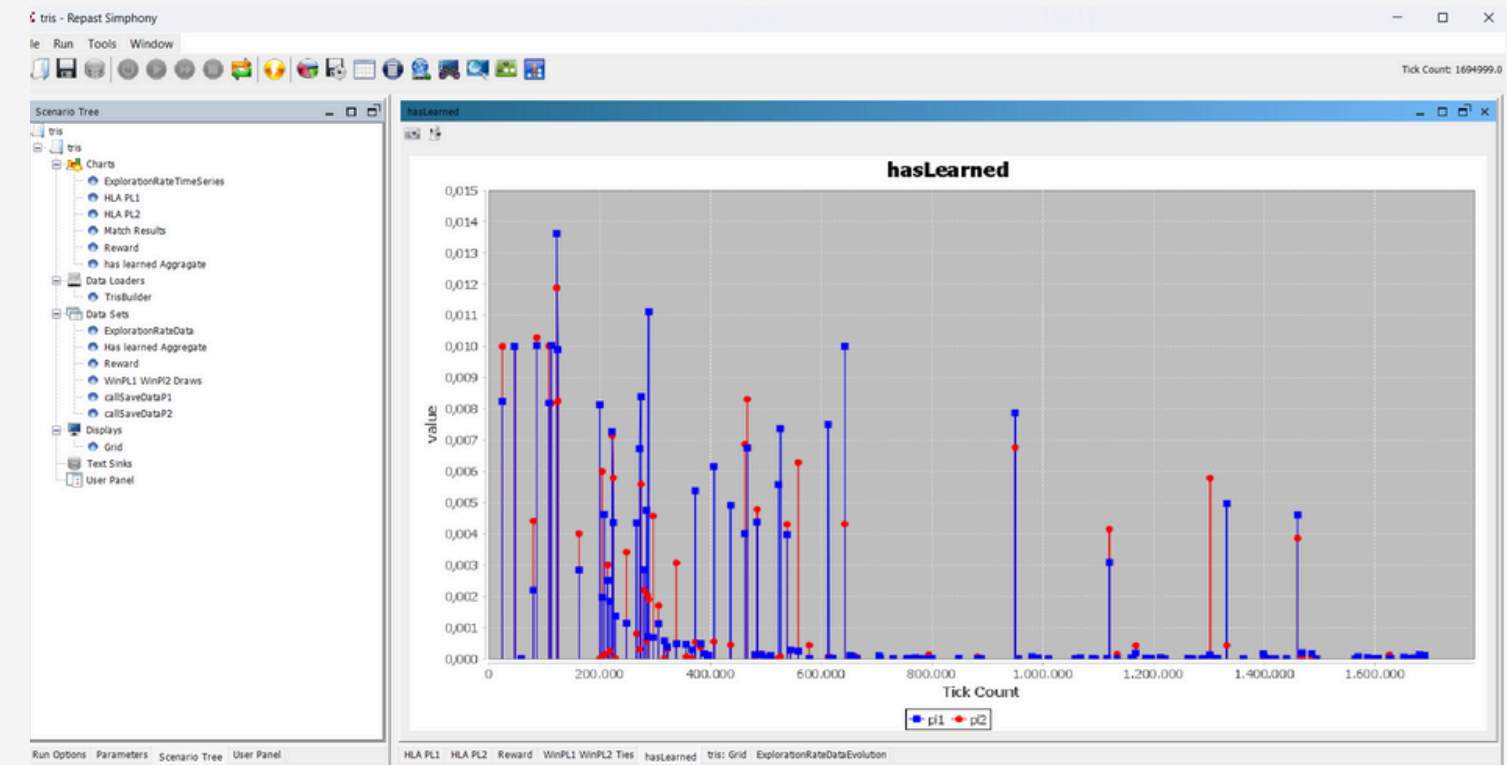
Griglia 3x3 con logica a
passo singolo e Epsilon con
decadimento



TIC-TAC-TOE

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Griglia 3x3 con logica a
passo doppio e Epsilon con
decadimento

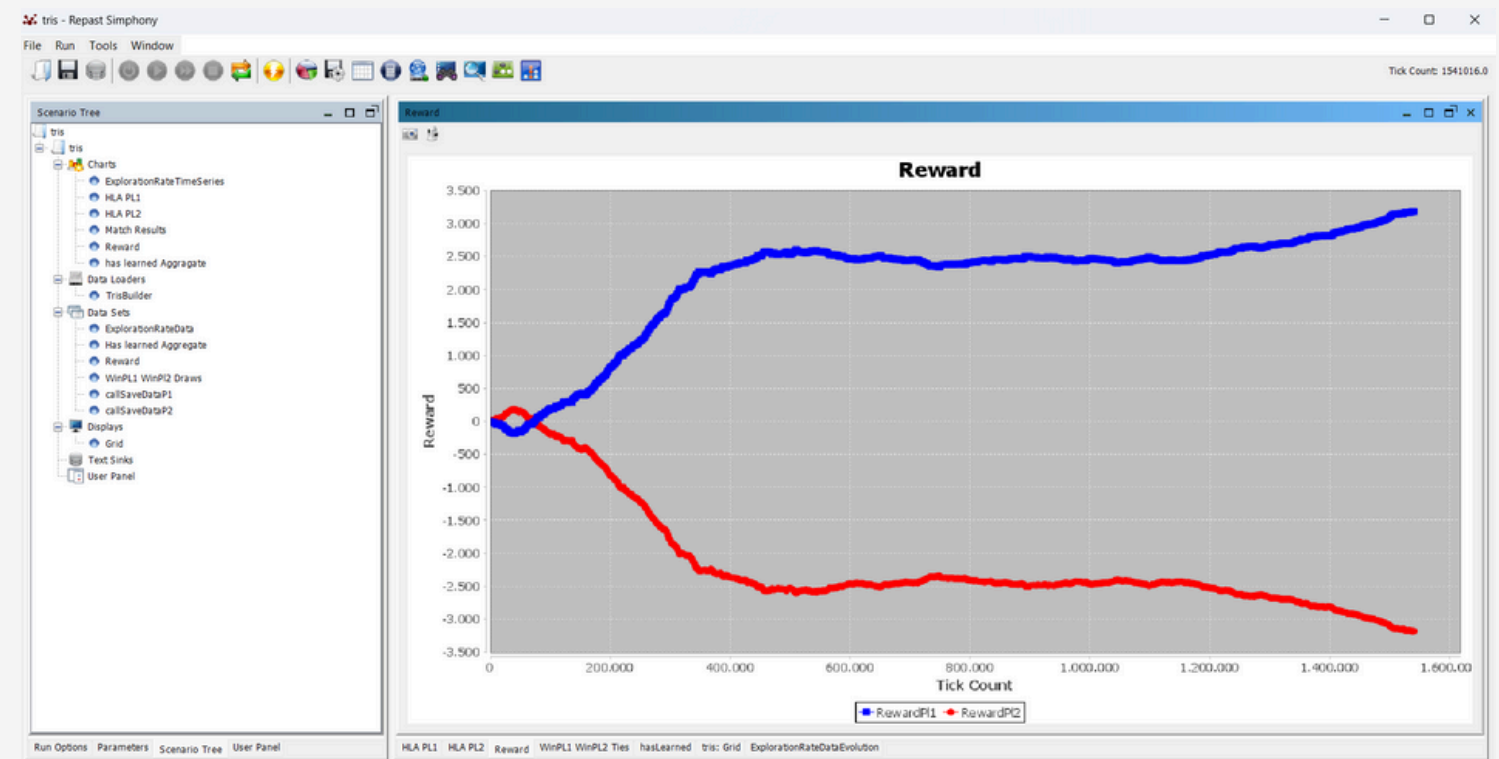
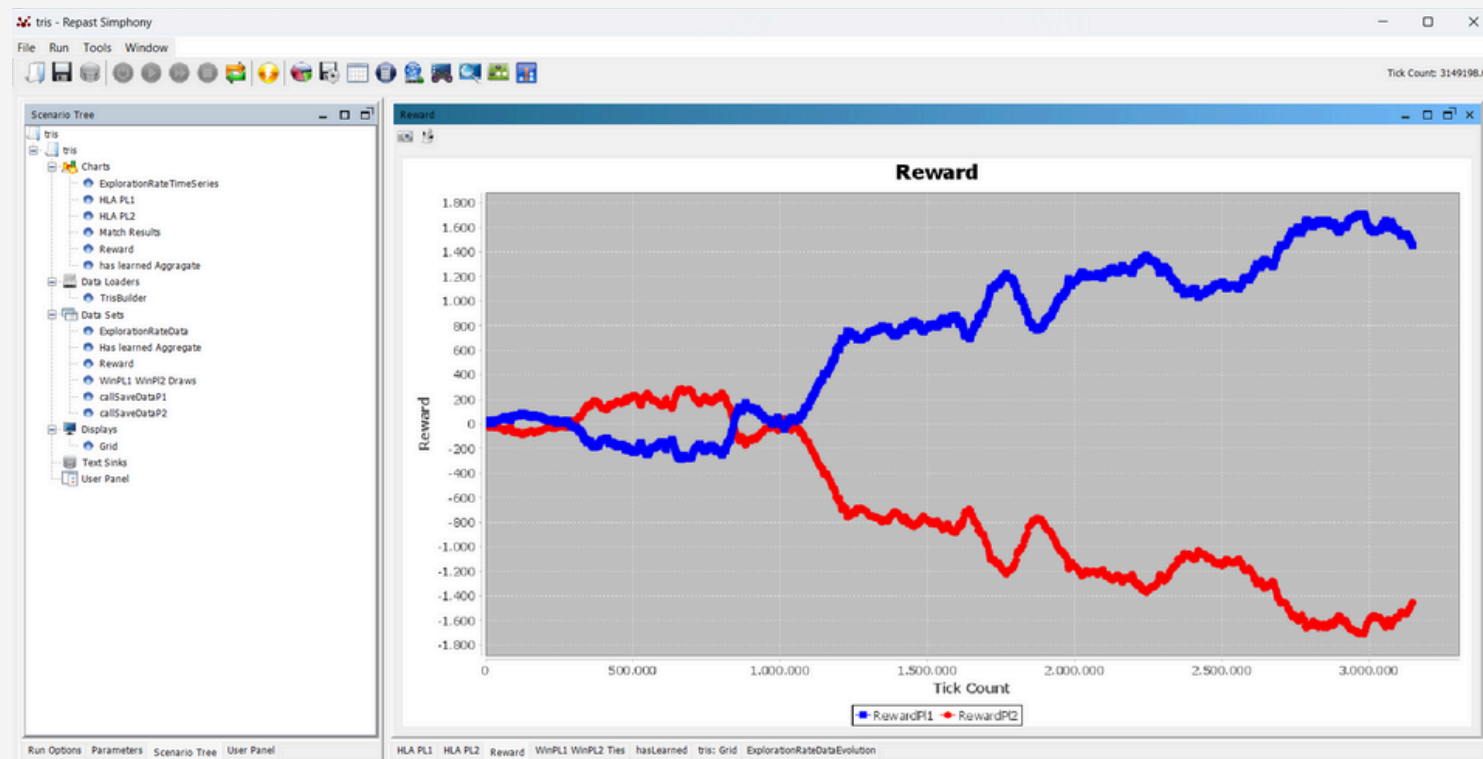
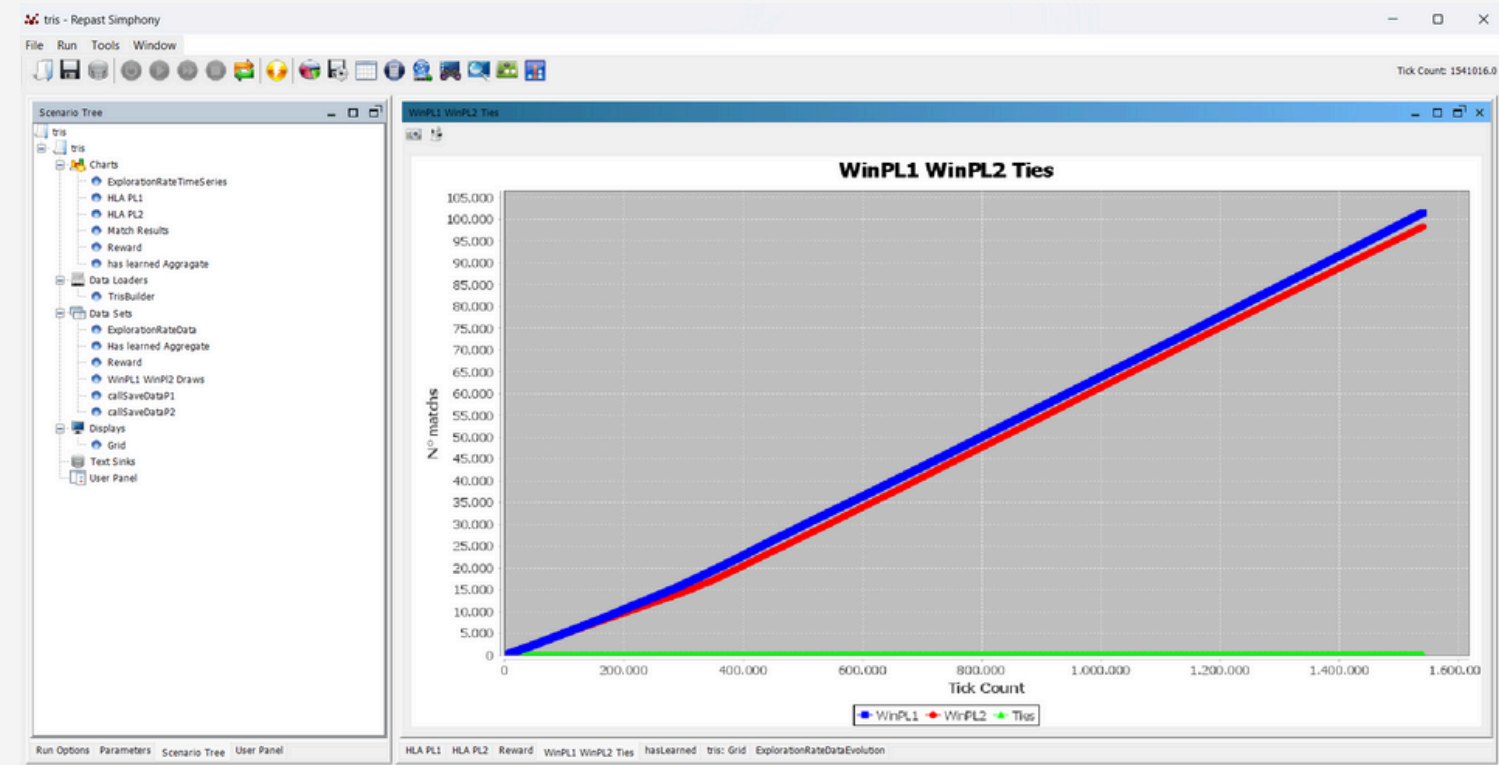


TIC-TAC-TOE

04

Griglia 4x4 con logica a
passo doppio

3 ELEMENTI IN FILA PER VINCERE

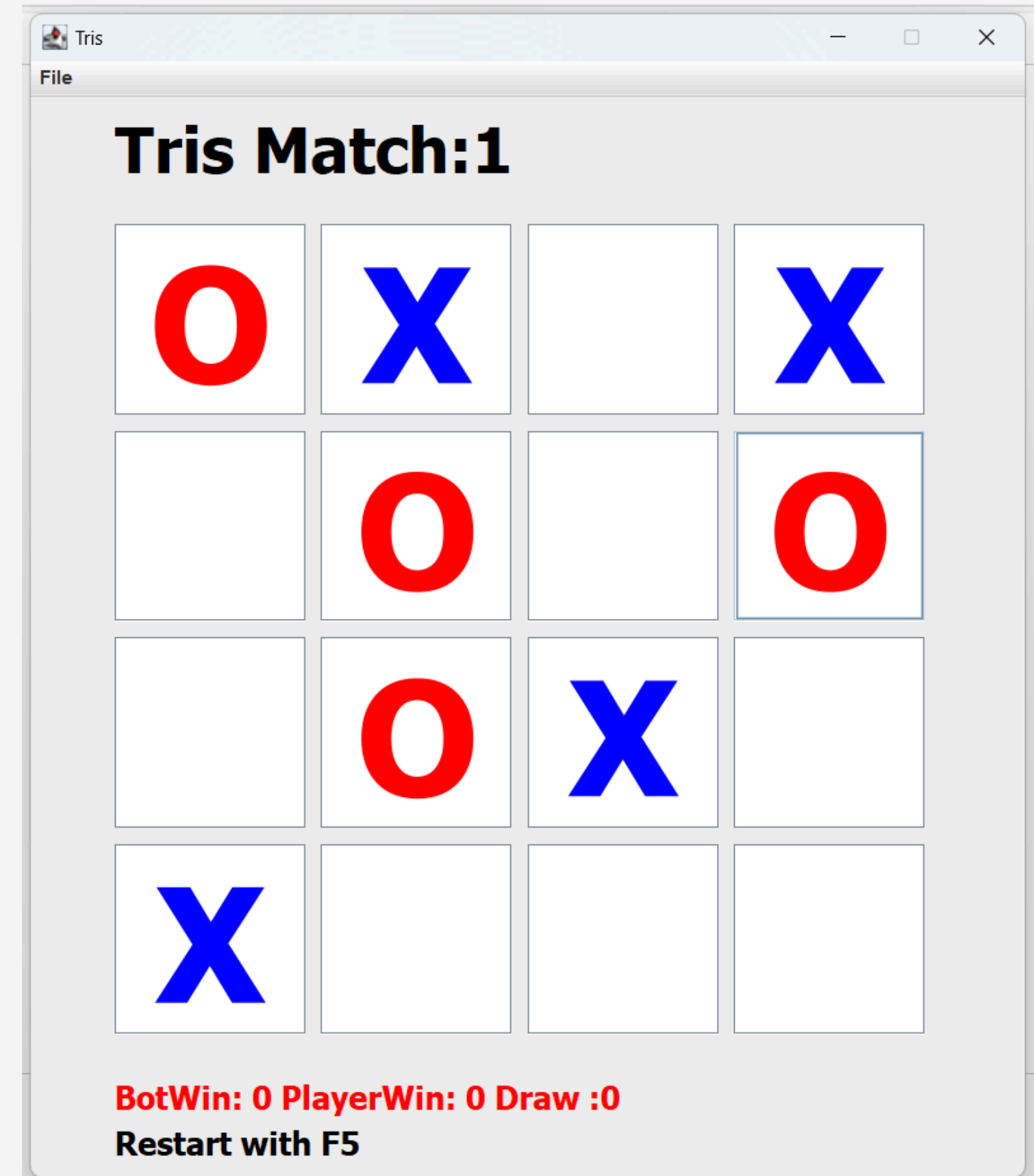
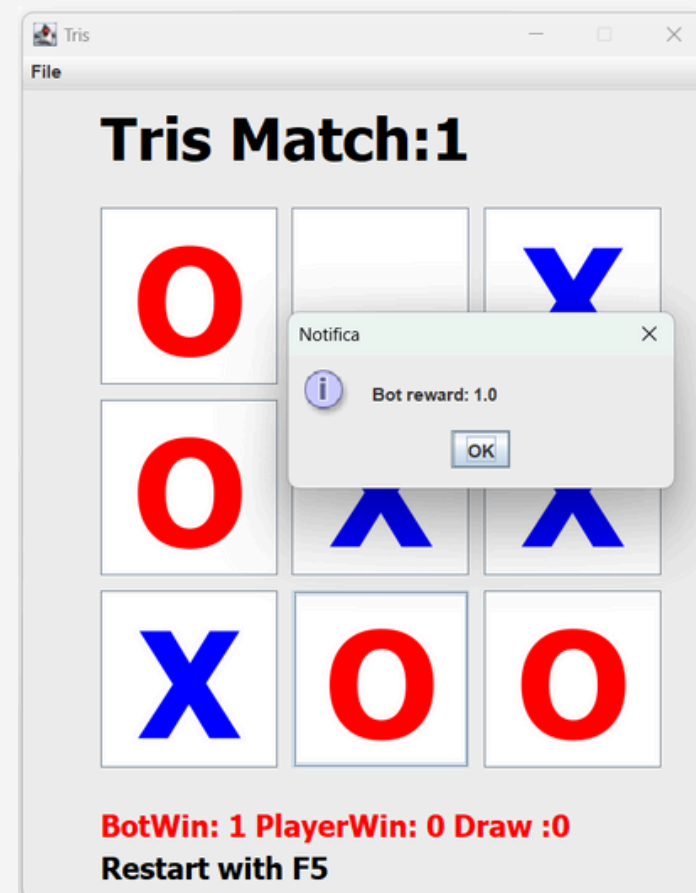


4 ELEMENTI IN FILA PER VINCERE

TIC-TAC-TOE

04

Interfaccia grafica
Giocatore umano
Vs Agente



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Grazie per l'attenzione

Anno Accademico 2023/24



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